## What is AQI?

AQI (Air Quality Index) is a single number that represents how polluted the air is and how it affects human health.

### Instead of showing many pollutants separately (PM2.5, PM10, CO, NO2, etc.), AQI:

Converts pollutant values into sub-indices

Takes the worst (maximum) sub-index

Reports it as the AQI of that area

## Why AQI Analysis is Important

Helps government issue health warnings

Helps citizens decide outdoor activities

Helps researchers study pollution trends

## STEP 1: Import Required Libraries

In [1]:
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

### Why are we using these?

pandas → For reading, cleaning, and analyzing data

matplotlib → For visualization (graphs & trends)

## STEP 2: Load the Dataset

In [2]:
```
df=pd.read_csv("delhiaqi.csv")
```

### What this does:

Reads the CSV file

Stores it as a DataFrame (table-like structure)

### Always Check Your Data

In [3]:
```
df.head()
```

Out[3]:

| | date | co | no | no2 | o3 | so2 | pm2_5 | pm10 | nh3 |
|---|---|---|---|---|---|---|---|---|---|
| **0** | 2023-01-01 00:00:00 | 1655.58 | 1.66 | 39.41 | 5.90 | 17.88 | 169.29 | 194.64 | 5.83 |
| **1** | 2023-01-01 01:00:00 | 1869.20 | 6.82 | 42.16 | 1.99 | 22.17 | 182.84 | 211.08 | 7.66 |
| **2** | 2023-01-01 02:00:00 | 2510.07 | 27.72 | 43.87 | 0.02 | 30.04 | 220.25 | 260.68 | 11.40 |
| **3** | 2023-01-01 03:00:00 | 3150.94 | 55.43 | 44.55 | 0.85 | 35.76 | 252.90 | 304.12 | 13.55 |
| **4** | 2023-01-01 04:00:00 | 3471.37 | 68.84 | 45.24 | 5.45 | 39.10 | 266.36 | 322.80 | 14.19 |

Why?

1. Confirms data loaded correctly
2. Shows column names and sample values

## STEP 3: Data Cleaning & Preprocessing

### 3.1 Convert Date Column

In [4]:
```
df['date'] = pd.to_datetime(df['date'])
```

Why?

. Date is initially a string

. Converting allows:

1. Time-based analysis
2. Monthly/daily trends

### 3.2 Check Missing Values

In [5]:
```
df.isnull().sum()
```

Out[5]:
```
date     0
co       0
no       0
no2      0
o3       0
so2      0
pm2_5    0
pm10     0
nh3      0
dtype: int64
```

***dataset has no missing values.***

## STEP 4: Overall pollution severity

### Average pollutant levels

In [6]:
```
pollutants = ["pm2_5", "pm10", "no2", "so2", "co", "o3"]
df[pollutants].describe()
```

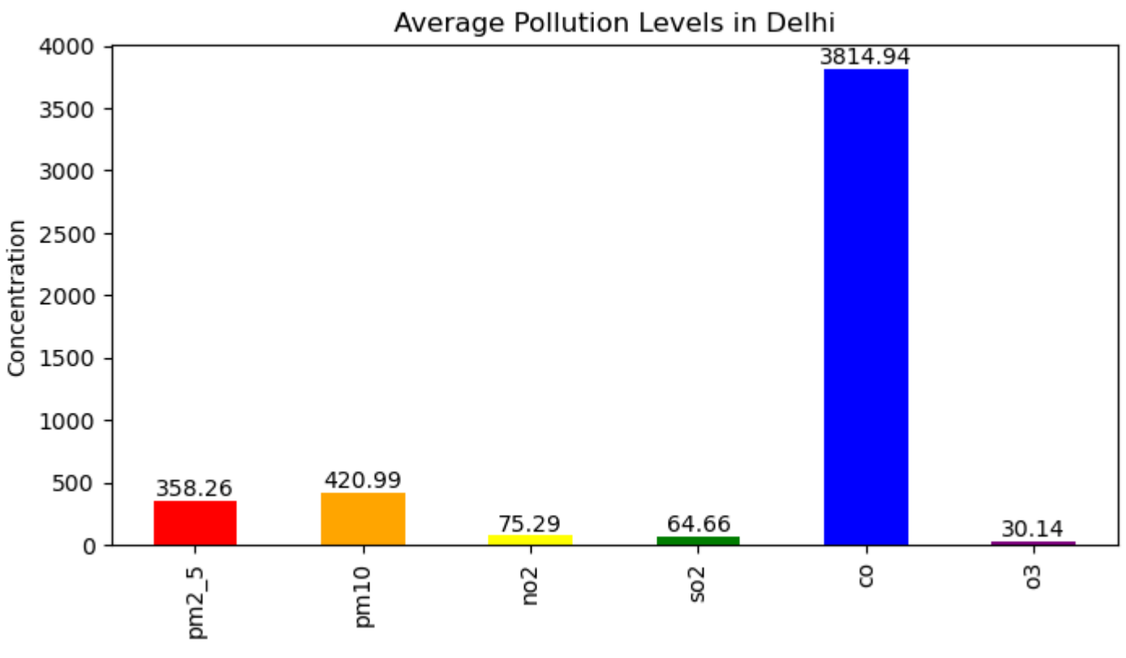
Out[6]:

| | pm2_5 | pm10 | no2 | so2 | co | o3 |
|---|---|---|---|---|---|---|
| **count** | 561.000000 | 561.000000 | 561.000000 | 561.000000 | 561.000000 | 561.000000 |
| **mean** | 358.256364 | 420.988414 | 75.292496 | 64.655936 | 3814.942210 | 30.141943 |
| **std** | 227.359117 | 271.287026 | 42.473791 | 61.073080 | 3227.744681 | 39.979405 |
| **min** | 60.100000 | 69.080000 | 13.370000 | 5.250000 | 654.220000 | 0.000000 |
| **25%** | 204.450000 | 240.900000 | 44.550000 | 28.130000 | 1708.980000 | 0.070000 |
| **50%** | 301.170000 | 340.900000 | 63.750000 | 47.210000 | 2590.180000 | 11.800000 |
| **75%** | 416.650000 | 482.570000 | 97.330000 | 77.250000 | 4432.680000 | 47.210000 |
| **max** | 1310.200000 | 1499.270000 | 263.210000 | 511.170000 | 16876.220000 | 164.510000 |

Which pollutants are worst?

**Mean comparison**

In [7]:
```
colors = ['red', 'orange', 'yellow', 'green', 'blue', 'purple']

ax = df[pollutants].mean().plot(
    kind='bar',
    figsize=(8,4),
    color=colors)
plt.title("Average Pollution Levels in Delhi")
plt.ylabel("Concentration")
# Add data labels on each bar--->
for container in ax.containers:
    ax.bar_label(container, fmt='%.2f')
plt.show()
```




***Seasonal analysis***

***Create seasons:***

In [8]:
```
df["month"] = df["date"].dt.month

def get_season(m):
    if m in [12,1,2]:
        return "Winter"
    elif m in [3,4,5]:
        return "Summer"
    elif m in [6,7,8,9]:
        return "Monsoon"
    else:
        return "Post-Monsoon"

df["season"] = df["month"].apply(get_season)
print(df["season"])
```

```
0      Winter
1      Winter
2      Winter
3      Winter
4      Winter
        ...
556    Winter
557    Winter
558    Winter
559    Winter
560    Winter
Name: season, Length: 561, dtype: object
```

## STEP 5:Seasonal pollutant trend

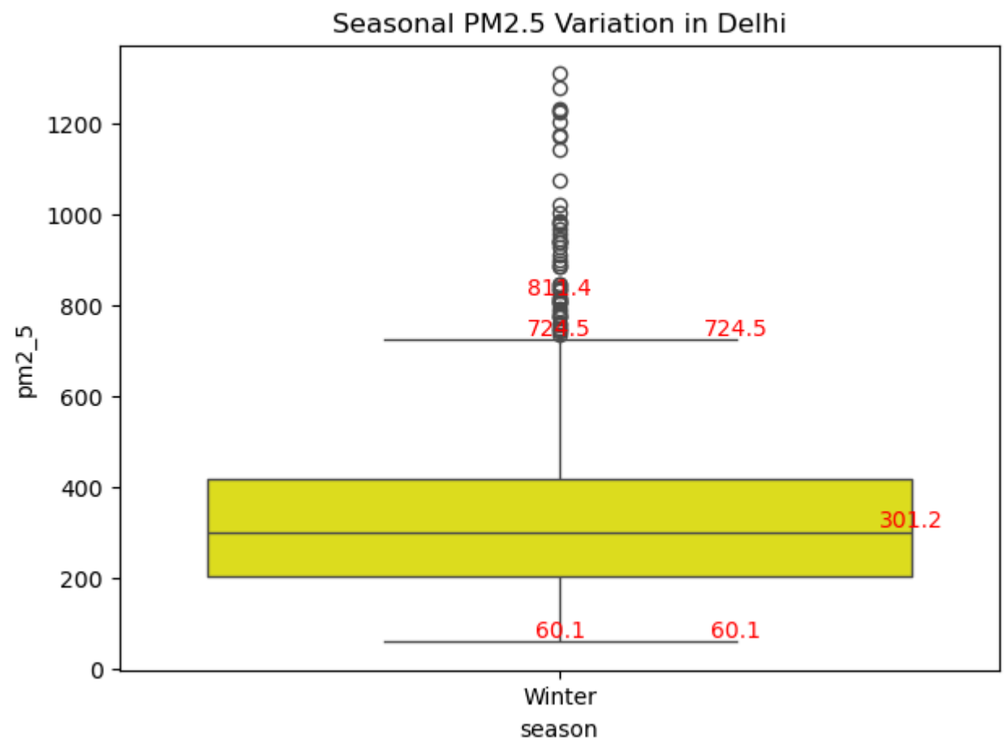
In [9]:
```
df.groupby("season")[pollutants].mean()
```

Out[9]:

| season | pm2_5 | pm10 | no2 | so2 | co | o3 |
|---|---|---|---|---|---|---|
| **Winter** | 358.256364 | 420.988414 | 75.292496 | 64.655936 | 3814.94221 | 30.141943 |

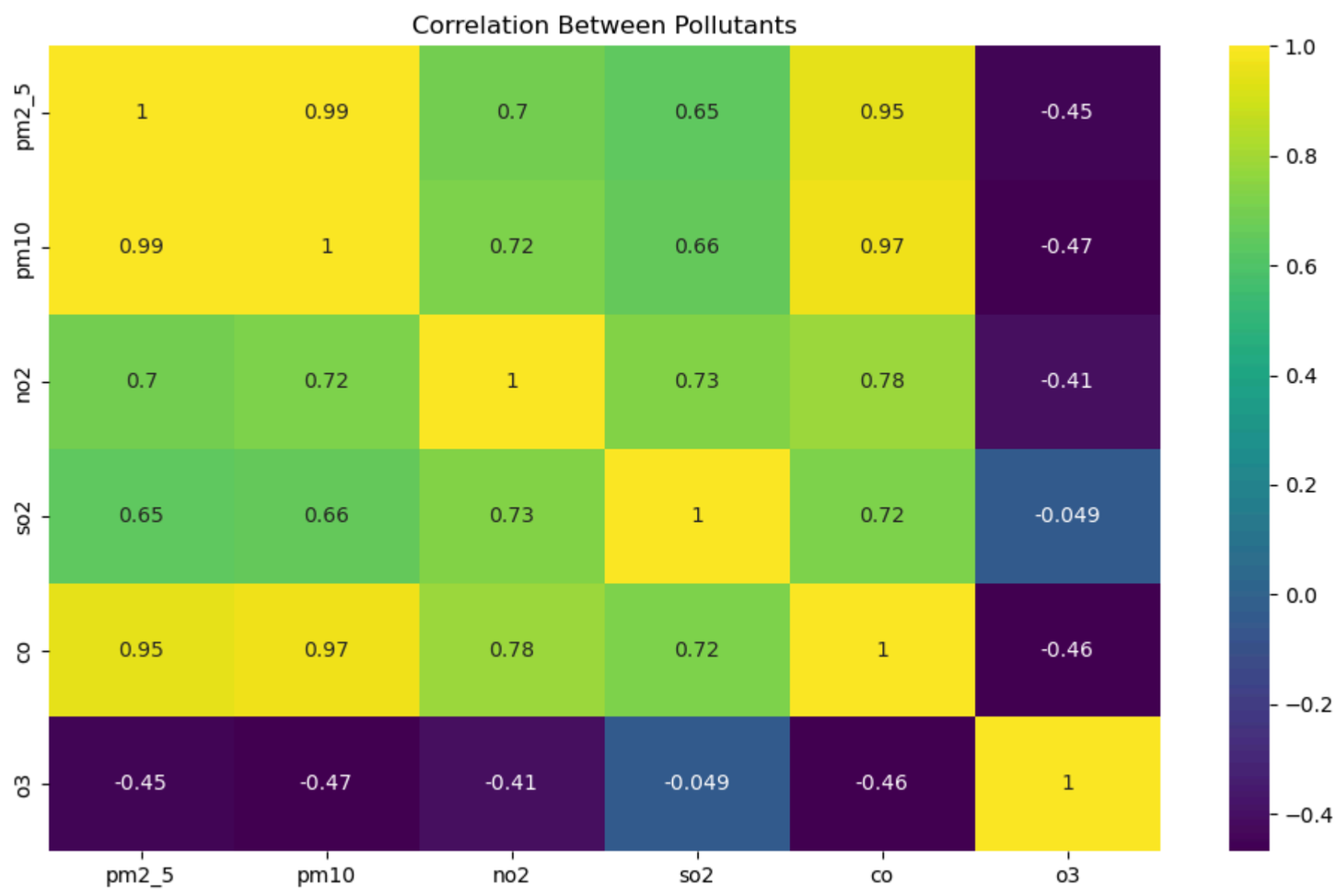
## STEP 6: Visualization:

In [21]:
```
plt.figure(figsize=(7,5))
ax=sns.boxplot(x="season", y="pm2_5", data=df,color="yellow")

## add median levels :
for line in ax.lines:
    x,y=line.get_xydata()[1]
    ax.text(x,y,f'{y:.1f}',ha="center",
            va="bottom",color="red")
plt.title("Seasonal PM2.5 Variation in Delhi")
plt.show()
```




STEP 7:Correlation between pollutants

In [11]:
```
corr = df[pollutants].corr()

plt.figure(figsize=(12,7))
sns.heatmap(corr, annot=True, cmap="viridis")
plt.title("Correlation Between Pollutants")
plt.show()
```




the maximum PM2.5 value

In [12]:
```
df["pm2_5"].max()
```

Out[12]:
```
1310.2
```

the date for it

In [13]:
```
df.loc[df["pm2_5"].idxmax(), ["date", "pm2_5"]]
```

Out[13]:
```
date     2023-01-19 17:00:00
pm2_5                 1310.2
Name: 449, dtype: object
```

Date with highest PM10

In [14]:
```
df.loc[df["pm10"].idxmax(), ["date", "pm10"]]
```

Out[14]:
```
date    2023-01-19 17:00:00
pm10                1499.27
Name: 449, dtype: object
```

Date with highest value for EACH pollutant

In [15]:
```
for p in pollutants:
    print(p)
    print(df.loc[df[p].idxmax(), ["date", p]])
    print("------")
```

```
pm2_5
date     2023-01-19 17:00:00
pm2_5                 1310.2
Name: 449, dtype: object
------
pm10
date    2023-01-19 17:00:00
pm10                1499.27
Name: 449, dtype: object
------
no2
date    2023-01-19 16:00:00
no2                  263.21
Name: 448, dtype: object
------
so2
date    2023-01-19 17:00:00
so2                  511.17
Name: 449, dtype: object
------
co
date    2023-01-13 17:00:00
co                16876.22
Name: 305, dtype: object
------
o3
date    2023-01-17 09:00:00
o3                   164.51
Name: 393, dtype: object
------
```

## Overall most polluted day

### Create pollution load

In [16]:
```
df["Total_Pollution"] = df[pollutants].sum(axis=1)
print(df["Total_Pollution"])
```

```
0      2082.70
1      2329.44
2      3064.93
3      3789.12
4      4150.32
        ...
556    2384.17
557    2357.07
558    2601.17
559    1943.79
560    1646.52
Name: Total_Pollution, Length: 561, dtype: float64
```

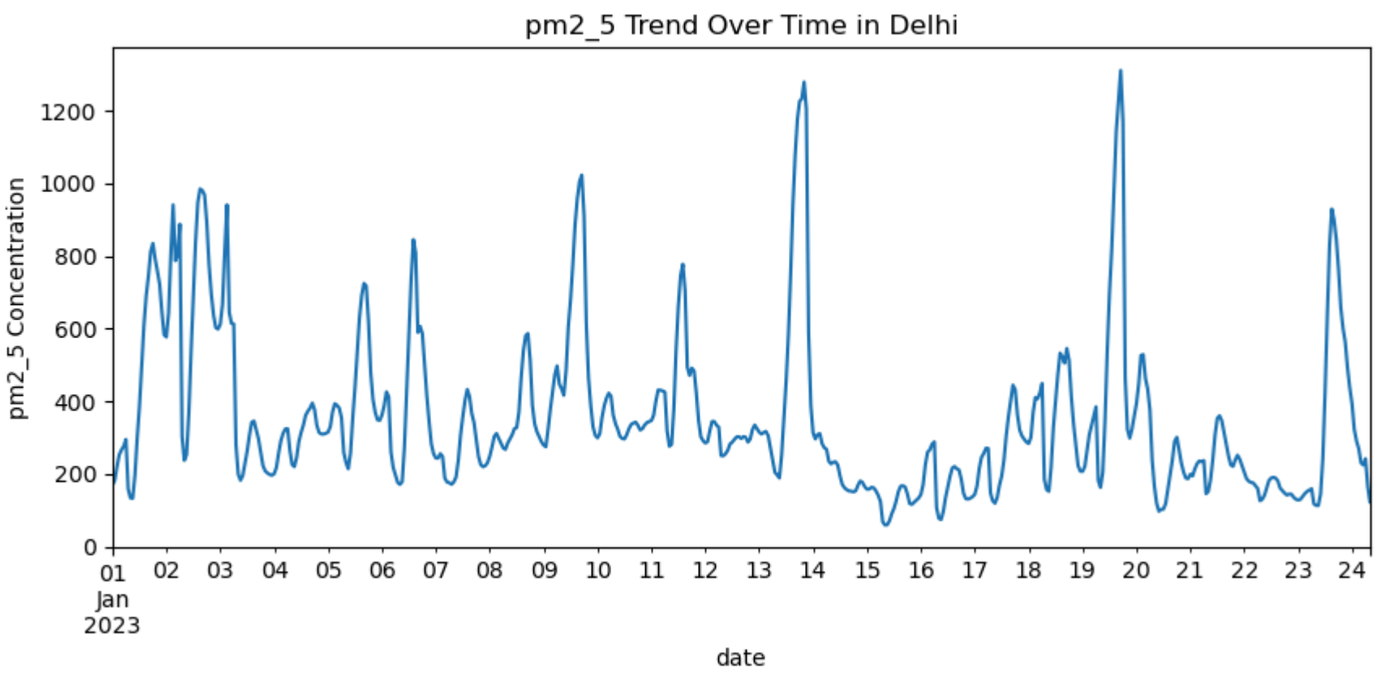
## Find worst day

In [17]:
```
df.loc[df["Total_Pollution"].idxmax(), ["date", "Total_Pollution"]]
```

Out[17]:
```
date               2023-01-19 17:00:00
Total_Pollution               20235.49
Name: 449, dtype: object
```

## Visualize pollution over time

In [18]:
```
df.set_index("date")["pm2_5"].plot(figsize=(10,4))
plt.title("pm2_5 Trend Over Time in Delhi")
plt.ylabel("pm2_5 Concentration")
plt.show()
```




Second video:***presentation video no. II***

In [ ]: